

Vaishak Menon

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WORK EXPERIENCE

LPL Financial

Fort Mill, South Carolina

Internal Audit - Data Analyst

Jun 2026 - Present

- Developing a **deterministic PDF parsing pipeline** to standardize document analysis workflows, reducing reliance on manual review.
- Automating legacy manual processes with **Power Automate**, replacing manual workflows with structured, repeatable flows.
- Refactoring **backend scripts** for maintainability and long-term scalability, reducing technical debt in legacy systems.

Speridian Technologies

Dillon, Montana

Associate Business Analyst

Jan 2026 - Jun 2026

- Facilitated **requirements gathering** sessions with key stakeholders to **define project scope**, ensuring all deliverables aligned with strategic business goals and minimized ambiguity.
- Developed detailed **user stories** and **acceptance criteria** for the product backlog, providing clear, actionable direction for development teams during **Agile sprint cycles**.
- Translated **business needs** into **technical specifications** and **process flows** for stakeholders and engineering teams.

Finchirp

Remote

Backend Software Developer

Aug 2025 - Dec 2025

- Architected real-time financial data integration system integrating **multiple third-party APIs** with caching layer and RESTful endpoints, enabling platform to deliver live market data to users **at scale** using Agile methodologies.
- Developed proprietary **multi-factor scoring algorithm** combining weighted data from multiple sources to identify market trends, with optimized query performance through **intelligent caching strategies**.
- Delivered full-stack features across **3,000+ LOC** in **2-week sprint cycles**, integrating multiple external APIs, designing database schemas, and managing migrations using Jira for project tracking and coordination.

EDUCATION

University of Alabama at Birmingham

Birmingham, Alabama

B.S in Computer Science

Graduation Date: Apr 2024

M.S in Computer Science

Graduation Date: May 2025

PROJECTS

Portfolio RAG Chatbot | Next.js 15, React 19, FastAPI, ChromaDB, Redis, Groq API

- Built a **real-time streaming** interface using **Next.js 15** and **Server-Sent Events (SSE)** to visualize "chain-of-thought" reasoning, backed by a secure **Edge Proxy** to mask API keys and ensure network resilience.
- Optimized retrieval via **hybrid semantic-lexical reranking** and **negative inference** detection, achieving **2.12s mean latency** and eliminating hallucinations for non-existent entities.

Pomodoro Focus | Next.js 15, React 19, TypeScript, PostgreSQL, Tailwind v4

- Developed a production productivity platform using **Next.js 15** and **React 19**, leveraging **React Server Components** for optimized rendering and **Tailwind CSS v4** for a responsive glassmorphism UI.
- Engineered a strict **Content Security Policy (CSP)** and custom **subdomain authentication** flow via **Clerk** to securely manage third-party media integrations and resolve cross-site cookie restrictions.

ShakGPT | Python, PyTorch, GQA, RoPE, SwiGLU, KV Cache

- Built a **350M-parameter decoder-only transformer** from scratch, implementing **grouped-query attention**, **rotary positional embeddings**, and **SwiGLU** activations for efficient long-context inference.
- Pretrained the model on **~6.7B tokens** across FineWeb, Wikipedia, Gutenberg, and StarCoder, implementing **KV caching** to optimize inference-time memory and latency during generation.

CERTIFICATIONS & SKILLS

Certifications: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, Alteryx Designer Core

Skills: Python, TypeScript, React, Next.js, Java, C, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, Terraform, GCP, AWS, Git